Yousuf Shahzad

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Education

University College London

BSc Computer Science — *Sep 2026 – Jun 2029*

- **Relevant Modules:** Principles of Programming, Algorithms, Object-Oriented Programming, Theory of Computation, and Discrete Mathematics for Computer Science

Upton Court Grammar School — Slough, UK

Secondary School & Sixth Form — *Sep 2019 – Jul 2026*

- **A-Levels:** Mathematics (A*), Further Mathematics (A*), Computer Science (A*); **EPQ:** A*
- **GCSEs:** 11 subjects at grades 9–7

Experience

Network Engineering Work Experience — Jul 2025

Virgin Media O2

- Worked alongside radio optimisation engineers on network planning, performance analysis and service delivery, exploring how network data informs engineering decisions

Technology and Cybersecurity Work Experience — Jun – Jul 2025

IBM

- Completed technical workshops on cybersecurity, AI security, cryptography, red teaming and penetration testing across technology and consulting teams

IT and Software Engineering Work Experience — Jul 2023

Europe Snacks

- Implemented an SQL-based feature in the SAP HANA ERP environment alongside the IT team; investigated software and support issues through Jira

Projects

Sting | *Accel AI Innovate London 2026*

- Placed **6th out of 42 teams**; contributed to the frontend, FastAPI backend and team presentation of a referral-promotion testing sandbox
- Implemented idempotent actions, SQLite/PostgreSQL persistence and independent financial checks with deterministic replay to reproduce simulated policy failures; added pytest coverage

Mathematics Society Web Lead | *Upton Court Grammar School*

- Built and operated a Flask/SQLAlchemy platform for a school community of 1,200+ students, supporting 150+ active accounts, challenge submissions and live leaderboards
- Deployed on a self-managed VPS with Nginx, DNS and Cloudflare; developed authentication, role-based administration and timed challenges, with pytest and GitHub Actions testing

VaultKey | *Local-Only Password Manager*

- Developed a Rust/Tauri and React/TypeScript desktop password manager with session-gated credential management, password generation and fuzzy search
- Encrypted local vaults with AES-256-GCM and Argon2id key derivation; wrote Rust unit tests for cryptography, authentication and credential-management flows

Nex | *Minecraft Server Management CLI*

- Built a Python CLI to provision and run Minecraft servers, integrating Modrinth/Spigot plugin installation, dependency checks, version pinning and Java compatibility validation

Additional Information

Technical Skills: Python, Rust, JavaScript/TypeScript, SQL, HTML/CSS; Flask, FastAPI, React, Tauri, SQLAlchemy, pytest; Git, GitHub Actions, PostgreSQL, Nginx, SAP HANA, Jira
Awards: 2nd place, Lockheed Martin Code Quest (Advanced, Ampthill, 2025); top 18% globally, MathWorks M3 Challenge; MEI Ritangle finalist; Duke of Edinburgh Silver and Bronze
Leadership: School Head Prefect, Careers Committee Head, Head of Computer Science Society
Interests: Motorsport, reading, running and self-hosted technology